
Algorithmen und Wahrscheinlichkeit

Exercise Session 7

Exercise 1 – *Max Cut*

Consider a graph $G = (V, E)$. For a subset $S \subset V$, we say that let $E(S, V \setminus S)$ be the set of all edges $\{u, v\}$ s.t. $u \in S$ and $v \in V \setminus S$ (we call this a *cut* of G , of size $|E(S, V \setminus S)|$). We will first show that there exist a cut of size $|E|/2$.

1. Consider a random set $S \subset V$, generated by including each $v \in V$ into S independently at random with probability $1/2$. For each edge $e \in E$, define $X_e \in \{0, 1\}$ to be the indicator of an event $e \in E(S, V \setminus S)$, and $X = |E(S, V \setminus S)|$. Show that

$$\mathbb{E}[X] = |E|/2,$$

and deduce that there exist a cut of G of size at least $|E|/2$.

2. We will see an algorithm to generate a set of size not much smaller. For two edges $e, f \in E$, compute $\mathbb{E}[X_e X_f]$ (hint: distinguish 3 cases, depending on whether e and f has 0, 1 or 2 vertices on common).
3. Show that $\mathbb{E}[X^2] = \mathbb{E}[X]^2$. Deduce that $\text{Var}[X] \leq \mathbb{E}[X]^2$.
4. Show that with probability $2/3$ if we sample random subset S , then the cut $E(S, V \setminus S)$ is of size at most $|E|/2 - C\sqrt{|E|}$ for some constant C .
5. Provide an algorithm that finds a cut of size $|E|/2 - C\sqrt{|E|}$ for the same constant C with probability at least $1 - \delta$ for a δ given as in input to the algorithm.